WORKED SOLUTION

STEP III 2021, Question 11

Probability and Statistics • Exponential Distribution

STEP III 2021, Question 11

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 12.18, *Memorylessness*: https://beyondstep.keithmcnulty.org/ch12/memoryless.html

The key idea

Every question about Y and Z is really a question about which interval X lands in. The event $Y = n$ is $n \le X < n+1$, and the event $Z < z$ splits into the disjoint pieces $n \le X < n+z$, one for each n. The exponential distribution's tail $\mathrm{P}(X \ge x) = \mathrm{e}^{-\lambda x}$ turns each piece into a term of a geometric series.

Solution

Throughout we write $q = \mathrm{e}^{-\lambda}$, so $0 < q < 1$ because $\lambda > 0$. For $x \ge 0$,

$$\mathrm{P}(X \ge x) = \int_x^\infty \lambda \mathrm{e}^{-\lambda t}\,\mathrm{d}t = \mathrm{e}^{-\lambda x},$$

and since X is continuous, $\mathrm{P}(a \le X < b) = \mathrm{P}(a < X < b) = \mathrm{e}^{-\lambda a} - \mathrm{e}^{-\lambda b}$ for $0 \le a < b$.

Part (i)

Y is the integer part of X, so $Y = n$ exactly when $n \le X < n+1$. Hence

$$\mathrm{P}(Y = n) = \mathrm{e}^{-n\lambda} - \mathrm{e}^{-(n+1)\lambda} = (1 - \mathrm{e}^{-\lambda})\mathrm{e}^{-n\lambda},$$

as required. (So Y is geometric, starting at 0, with ratio q.)

Part (ii)

$Z = X - Y$ is the fractional part of X, so $0 \le Z < 1$. Take $0 \le z \le 1$. The event $Z < z$ happens exactly when, for the integer $n = Y$, we have $n \le X < n + z$. These events for

$n = 0, 1, 2, \ldots$ are disjoint, so

$$\begin{aligned} \mathrm{P}(Z < z) &= \sum_{n=0}^{\infty} \mathrm{P}(n \le X < n + z) = \sum_{n=0}^{\infty} \left(\mathrm{e}^{-n\lambda} - \mathrm{e}^{-n\lambda - \lambda z}\right) \\ &= (1 - \mathrm{e}^{-\lambda z}) \sum_{n=0}^{\infty} q^n = \frac{1 - \mathrm{e}^{-\lambda z}}{1 - \mathrm{e}^{-\lambda}}, \end{aligned}$$

summing the geometric series, which converges because $0 < q < 1$. This is the required result.

Part (iii)

Differentiating the distribution function from (ii), Z has density

$$g(z) = \frac{\lambda \mathrm{e}^{-\lambda z}}{1 - \mathrm{e}^{-\lambda}} \qquad (0 \le z \le 1),$$

and $g(z) = 0$ otherwise. Integrating by parts with $u = z$ and $\mathrm{d}v = \lambda \mathrm{e}^{-\lambda z}\,\mathrm{d}z$,

$$\int_0^1 z\lambda \mathrm{e}^{-\lambda z}\,\mathrm{d}z = \left[-z\mathrm{e}^{-\lambda z}\right]_0^1 + \int_0^1 \mathrm{e}^{-\lambda z}\,\mathrm{d}z = -\mathrm{e}^{-\lambda} + \frac{1 - \mathrm{e}^{-\lambda}}{\lambda}.$$

Dividing by $1 - \mathrm{e}^{-\lambda}$,

$$\mathrm{E}(Z) = \frac{1}{\lambda} - \frac{\mathrm{e}^{-\lambda}}{1 - \mathrm{e}^{-\lambda}} = \frac{1}{\lambda} - \frac{1}{\mathrm{e}^{\lambda} - 1}.$$

Part (iv)

Let $0 \le z_1 < z_2 \le 1$. The event $\{Y = n \text{ and } z_1 < Z < z_2\}$ is exactly $n + z_1 < X < n + z_2$, so

$$\mathrm{P}(Y = n \text{ and } z_1 < Z < z_2) = \mathrm{e}^{-\lambda(n+z_1)} - \mathrm{e}^{-\lambda(n+z_2)} = \mathrm{e}^{-n\lambda}\left(\mathrm{e}^{-\lambda z_1} - \mathrm{e}^{-\lambda z_2}\right).$$

> **Ask yourself** What would we need to compare this with to decide about independence?

With the product $\mathrm{P}(Y = n)\,\mathrm{P}(z_1 < Z < z_2)$. Since Z is continuous, $\mathrm{P}(Z = z_1) = 0$, so by (ii)

$$\mathrm{P}(z_1 < Z < z_2) = \mathrm{P}(Z < z_2) - \mathrm{P}(Z < z_1) = \frac{\mathrm{e}^{-\lambda z_1} - \mathrm{e}^{-\lambda z_2}}{1 - \mathrm{e}^{-\lambda}}.$$

Multiplying by $\mathrm{P}(Y = n) = (1 - \mathrm{e}^{-\lambda})\mathrm{e}^{-n\lambda}$ from (i), the factors $1 - \mathrm{e}^{-\lambda}$ cancel and we get exactly the joint probability above. So

$$\mathrm{P}(Y = n \text{ and } z_1 < Z < z_2) = \mathrm{P}(Y = n)\,\mathrm{P}(z_1 < Z < z_2)$$

for every non-negative integer n and every interval $(z_1, z_2) \subseteq [0, 1]$. Since Y only takes non-negative integer values and Z only takes values in $[0, 1)$, events of these forms determine the whole joint distribution. Therefore Y and Z *are* independent: knowing which unit interval X fell in tells us nothing about where in that interval it fell.

Checks

- Putting $z = 1$ in (ii) gives $\mathrm{P}(Z < 1) = 1$, as it must, and $z = 0$ gives 0.
- As $\lambda \to 0$ the density of X is almost flat over each unit interval, so Z should be close to uniform on $[0, 1]$. Indeed $\mathrm{e}^{\lambda} - 1 = \lambda + \frac{1}{2}\lambda^2 + \cdots$ gives $\frac{1}{\mathrm{e}^{\lambda}-1} = \frac{1}{\lambda} - \frac{1}{2} + O(\lambda)$, so $\mathrm{E}(Z) \to \frac{1}{2}$. As $\lambda \to \infty$, X is almost always tiny and $\mathrm{E}(Z) \to 0$.
- For $\lambda = 1$, $\mathrm{E}(Z) = 1 - \frac{1}{\mathrm{e}-1} \approx 0.4180$, which matches a computer simulation of 200 000 values of X; the simulated joint frequencies in (iv) also matched the products of the marginals.

Common slips

- Writing $\mathrm{P}(Z < z) = \mathrm{P}(X < z)$, forgetting that $Z < z$ can happen in every unit interval, not just the first.
- Concluding independence from a single choice of n, z_1, z_2: the factorisation must hold for all of them.

Beyond STEP

The independence in (iv) is a shadow of the *memoryless property* of the exponential distribution: given $X \geq n$, the excess $X - n$ is again exponential with the same rate, whatever n is. This is why the exponential distribution models waiting times in Poisson processes, and why its integer part is geometric, the discrete memoryless distribution. Grimmett and Welsh, *Probability: An Introduction*, develops both distributions and the Poisson process.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.